



Monitoring, Evaluation, Learning, and Impact Assessment (MELIA) Plan

Digital Transformation Accelerator (DTA)

June 2025

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1. Accelerator overview and results framework

Accelerator Overview

The Digital Transformation Accelerator (DTA) will support CGIAR and its partners in co-designing inclusive, impactful digital solutions. By leveraging cutting-edge advancements in artificial intelligence (AI), machine learning, modeling, and big data analytics, the Accelerator aims to strengthen decision-making, enable smarter investments and policies, and drive widespread adoption of digital innovations, supporting CGIAR's five Impact Areas: nutrition and food security, poverty reduction, gender equality, environmental health, and climate resilience.

The Accelerator's approach is structured across four interconnected **Areas of Work (AoWs)**:

- **AoW 1 – Data Ecosystems** will address key challenges in data access and interoperability by strengthening cross-Center data sharing, implementing FAIR (Findable, Accessible, Interoperable, Reusable) principles, and advancing inclusive governance and collaborative analytics.
- **AoW 2 – Action Lab** will co-develop and accelerate digital solutions in priority use cases identified through stakeholder engagement. It emphasizes inclusivity via citizen science, gender and social inclusion, and human-centered design approaches.
- **AoW 3 – Digital Futures** will produce foresight case studies exploring the potential of emerging digital technologies while proactively identifying and addressing risks such as growing digital divides and ethical concerns.
- **AoW 4 – Enabling Environment** will create decentralized Digital Innovation Hubs that provide resources, training, and expertise. This workstream will also lead on ethics, data governance, behavioral change, and scaling models.

Use cases have been identified through listening sessions across 14 countries in the Global South, focusing on AI models to enhance access to CGIAR's data and knowledge products, digital tools for agronomy and crop breeding, youth engagement, and data-to-insight workflows. Achieving success requires strong institutional support, dedicated resources, technical talent, and partnerships. CGIAR will work with strategic partners like the Bill & Melinda Gates Foundation, Advanced Technology and Research Council (ATRC), and Google.org to leverage complementary strengths. Through the Accelerator, CGIAR aims to emerge as a global leader in responsible digital innovation—delivering smarter, faster, and more inclusive solutions for sustainable development.

The Theory of change

Narrative

The digital transformation in food, land, and water (FLW) systems requires empowering scientists, innovators, Science programs, CG Centers, and other stakeholders with the appropriate capacities, novel tools, and enabling environments across the digital ecosystem. When these actors have access to FAIR (Findable, Accessible, Interoperable, and Reusable), AI-ready data, user-friendly platforms, and advanced AI-driven novel digital tools, they will be able to generate insights more effectively, design innovations, and support in decision-making. To support this with targeted training and capacity development on responsible, inclusive digital innovation ensures that solutions are not only technically sound but also socially relevant and ethical.

The Accelerator will operate through four Areas of Work (AoWs): **Data Ecosystems**, **Action Lab**, **Digital Futures**, and **Enabling Environment**. A strong and functional digital ecosystem, cross-

sectoral partnerships among the scientists, partners, stakeholders, and institutional governance frameworks further enable the development and scaling of innovations that sustainably accelerate scientific research. These enabling environments support the creation of inclusive digital innovation networks and responsible data ecosystems that allow for equitable access, knowledge sharing, and collaborative problem-solving. As a result, scientists and innovators are better equipped to develop, deploy, and scale digital solutions that meet the real-world needs of the communities and systems they serve. These innovations help to accelerate science work, enhance research productivity, and promote equitable digital access across diverse contexts. The Digital Transformation Accelerator, defines its Theory of Change around four interconnected 2030 Outcomes: (1) Institutionalization of FAIR Data Principles, where CGIAR centers and partners fully adopt and apply FAIR (Findable, Accessible, Interoperable, and Reusable) practices; (2) Functional Partnerships for Digital Innovation, enabling co-creation through collaborative development and deployment of digital tools; (3) Widespread Use of DTA Innovations, reflecting broad uptake of DTA-provided solutions; and (4) Equitable Access and Adoption, ensuring that all users— especially smallholder farmers, women, and youth—can access and benefit from digital innovations.

These outcomes are underpinned by Intermediate Outcome (I-OCs), which mark key shifts in capabilities and practices. These include generating FAIR and AI-ready data, deploying open and secure data ecosystems, enhancing partner capacity for responsible digital use, and adopting governance frameworks that foster data sharing and innovation. AI-powered tools that boost research, decision-making, and collaboration, along with inclusive, user-centered digital services, directly support innovation use and equity goals. Other I-OCs strengthen awareness of frontier technologies, enable inclusive solution design, promote responsible innovation practices, and encourage policy environments that support digital transformation. Considering the rapidly changing landscape of LLMs and Generative AI, this I-OC will endow CGIAR and partner with the institutional capacity to deal with and adapt to these disruptive changes and be better positioned to harness these technologies.

To deliver these I-OCs, DTA provides a suite of High-Level Outputs (HLOs) including co-developed metadata standards, FAIR-compliant data tools, scalable analytics infrastructure, AI-enabled research tools, and training programs. These outputs also include frameworks for ethical AI, innovation hubs, digital business models, and knowledge products, which collectively ensure that innovations are responsibly developed, adopted, and scaled. This structured progression from outputs to intermediate outcomes leads directly to DTA's long-term vision of inclusive, data-driven, and sustainable transformation in agri-food systems.

Theory of Change Statement:

If, scientists, innovators and research partners have access to FAIR, AI-ready data platforms and are and equipped with novel digital tools, **And**

If, they receive targeted training and capacity support for responsible and inclusive digital innovation, **And**

If, enabling environments, partnerships, and governance systems are in place,

Then, they will be empowered to deploy ethical, inclusive, and scalable digital innovations that accelerate science and research programs, promote equitable access, and support sustainable development across food, land, and water systems.

Revision of ToC based on ISDC feedback

Following feedback from the Independent Science for Development Council (ISDC), the team conducted a comprehensive review of the ToCs for all four AoWs. ISDC's comments highlighted three key areas for improvement:

- Strengthening the linkage between High-Level Outputs (HLOs) and Intermediate Outcomes (I-OCs);
- Making the ToCs more explicit, particularly in identifying and integrating associated risks; and
- Providing greater clarity on delivery pathways and methodologies.

In response, the revised ToCs for all AoWs now address these concerns ([find here](#)). The linkages between HLOs and I-OCs have been made more explicit, with a clear reflection of how outputs contribute to outcomes and thereby 2030 outcomes. Additionally, each Area of Work (AoW) ToC includes key assumptions and associated risks that underpin the pathways to impact. Specific products and delivery methodologies have also been clearly outlined to enhance the transparency and operational clarity of each pathway. The board is aligned with these changes to communicate with the audience.

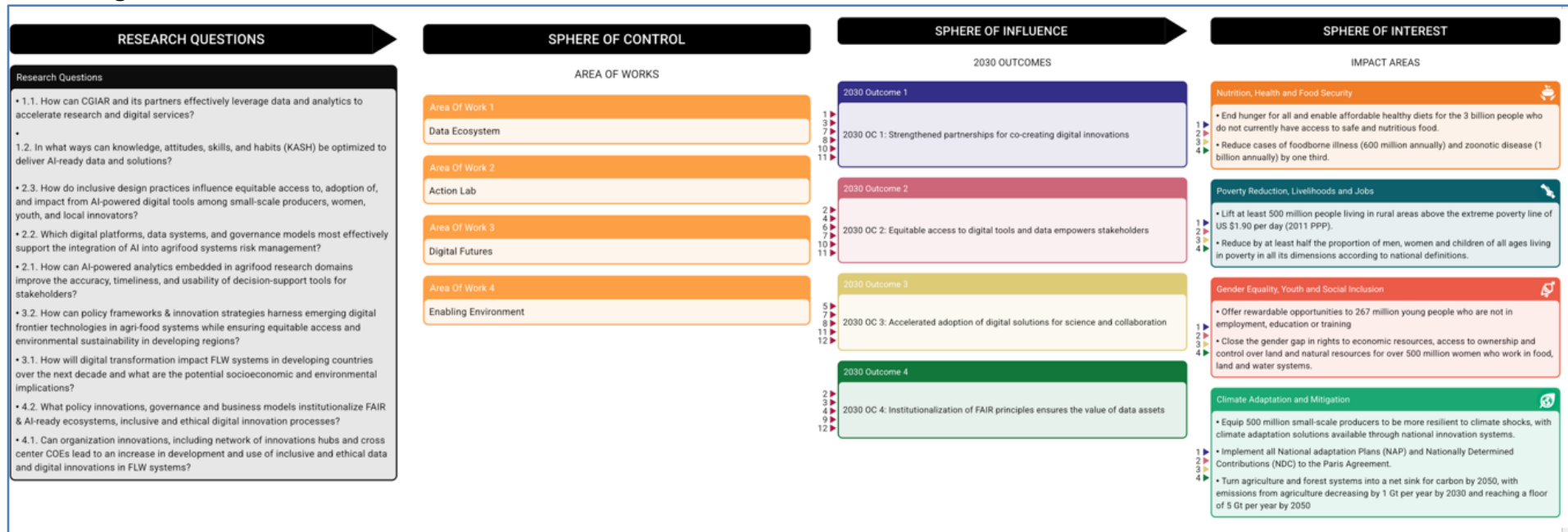
It is important to mention that as the Accelerator begins its implementation, ground-truth insights and learning from delivery processes will inform future iterations of the ToCs, allowing them to evolve and remain responsive to emerging evidence and operational realities.

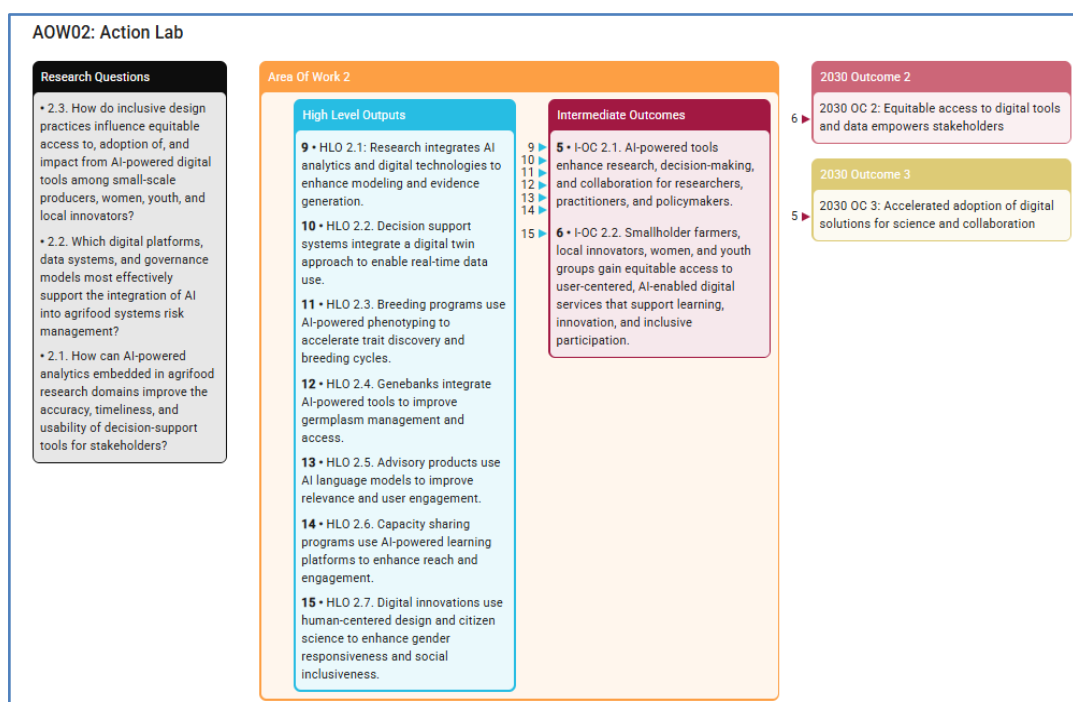
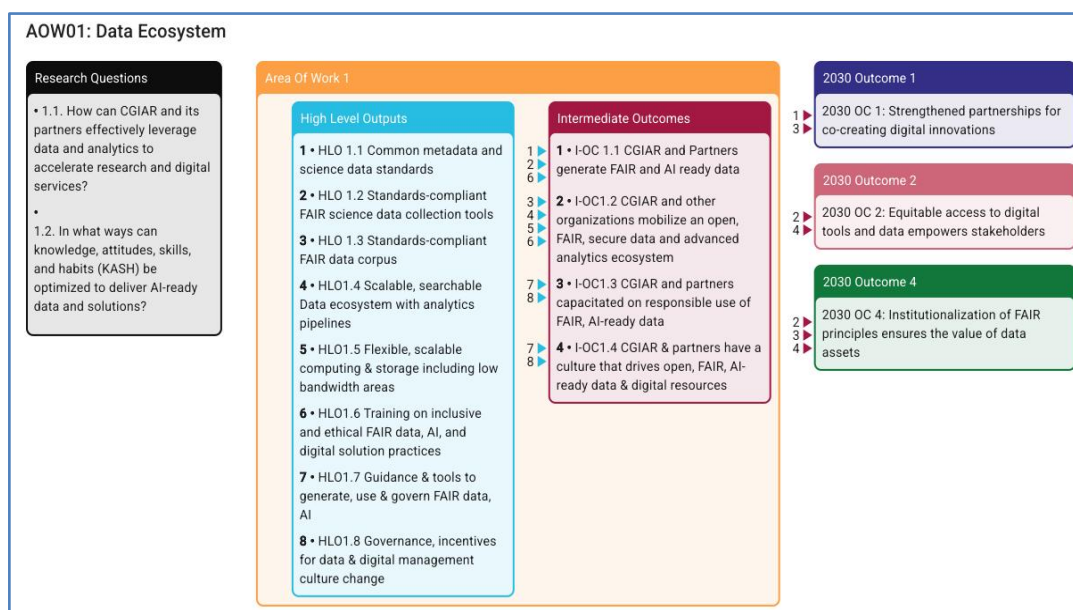
Key Assumptions:

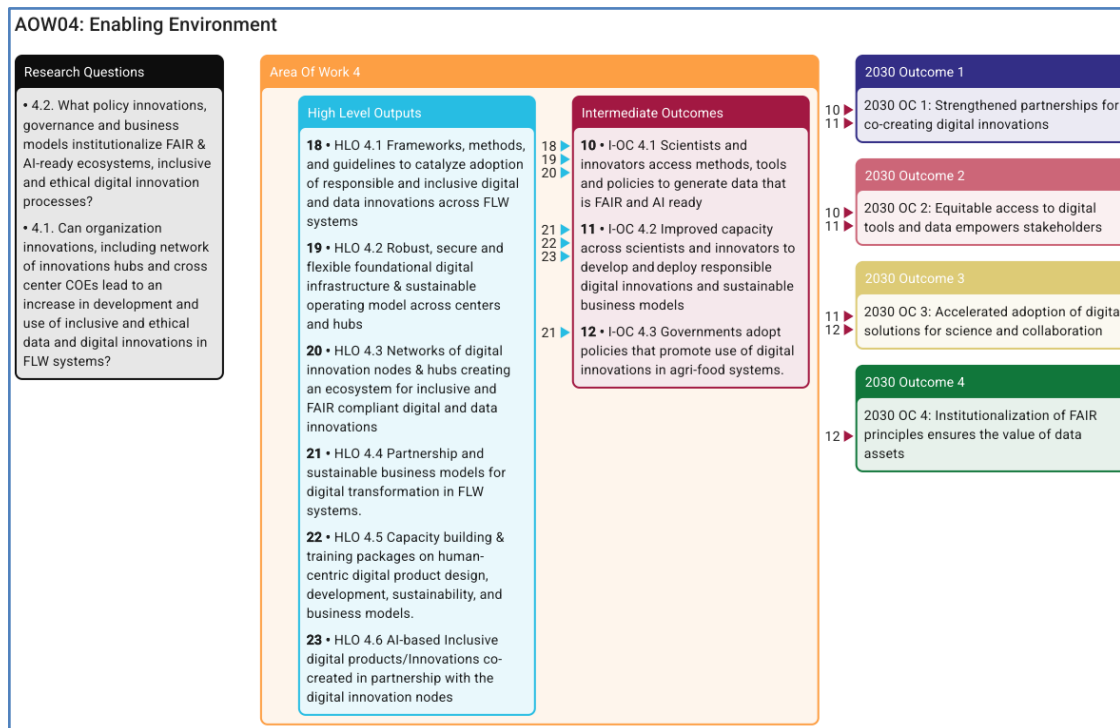
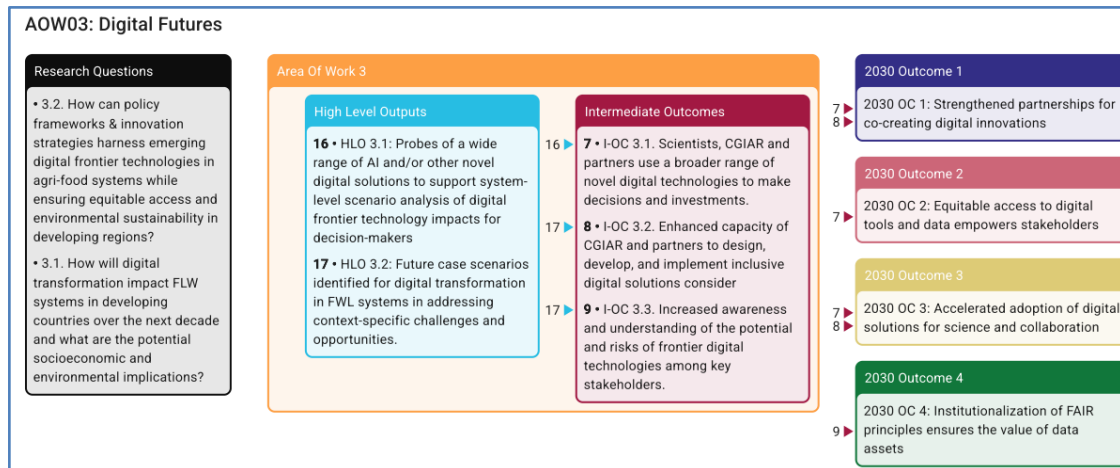
Like other programs, the DTA also relies on several key assumptions—such as the availability of strong partnerships, sustained investment, willingness to adopt innovations, and the presence of enabling ecosystems that support their adoption and use in research. The table below outlines these key assumptions and links them to the relevant Intermediate Outcome (I-OCs).

Key Assumption	Affected HLOs / I-OCs
Assumption 1: Stakeholders are willing and able to adopt FAIR principles and novel digital innovation	I-OC 1.1, I-OC 1.2, I-OC 2.1, I-OC 2.2, I-OC 4.3
Assumption 2: There is sufficient investment and infrastructure for secure, scalable data systems and digital innovation	I-OC 1.2, I-OC 1.4, I-OC 3.1, I-OC 3.2, I-OC 4.1,
Assumption 3: AI tools are aligned with research priorities and regulatory requirements	I-OC 1.3
Assumption 4: Clear governance, data ownership, and consent frameworks are in place	I-OC 3.3
Assumption 5: Ethical guidelines are followed to ensure trust in digital and AI systems	I-OC 3.2, I-OC 4.2
Assumption 6: Decision-makers are capable and willing to use AI-driven insights in policy	I-OC 2.1, I-OC 4.3
Assumption 7: Partnerships are strong and aligned under consistent governance frameworks	I-OC 3.3, I-OC 4.3

ToC diagrams- Accelerator Level







Results Framework

Please refer to the Results Framework attachment

AOW Title	Result type (Intermediate)	Result Short title	Indicator	Unit of measurement	Data source	Data collection method	Data collection frequency	Location	2025 – 2030	Target						2025 – 2030
										2025	2026	2027	2028	2029	2030	
AOW01 Data Ecosystem	High Level Output	HLO 1.1 Common metadata and science data standards	Number of standardized metadata and data guidelines developed and published across key research	Numbers	Report	Report review	Annually	global	0	2	2	2	2	2	2	10
AOW01 Data Ecosystem	High Level Output	HLO 1.2 Standards-compliant FAIR science data collection tools	Number of digital tools developed or adapted to collect FAIR-compliant data with standardized metadata and interoperability.	Numbers	Report	Report review	Annually	global	0	2	2	2	2	2	2	10
AOW01 Data Ecosystem	High Level Output	HLO 1.3 Standards-compliant FAIR data corpus	Number of datasets curated or produced under FAIR and standard-compliant protocols in scientific domains.	Number	Reports	Report review	Annually	global	0	5	10	10	10	10	5	50
AOW01 Data Ecosystem	High Level Output	HLO 1.4 Scalable, searchable Data ecosystem with analytics pipelines	Number of operational platforms enabling access, search, and analytics of scientific datasets.	Number	Report	Report review	Annually	global	0	1	1	1	1	1	1	6
AOW01 Data Ecosystem	High Level Output	HLO 1.5 Flexible, scalable computing & storage including low bandwidth areas	Number of infrastructure setups deployed for data storage and processing in target domains.	Number	Report	Report review	Annually	global	0	1	1	1	1	1	1	3
AOW01 Data Ecosystem	High Level Output	HLO 1.6 Training on inclusive and ethical FAIR data, AI, and digital solution practices	Number of individuals trained on ethical and inclusive practices for FAIR data and AI in scientific domains.	Number of individuals	Reports	Report review	Annually	global	0	200	200	200	200	200	200	1000
AOW01 Data Ecosystem	High Level Output	HLO 1.7 Guidance & tools to generate, use & govern FAIR data, AI	Number of guidance developed to generate, use & govern FAIR data, AI	Number	Report	Report review	Annually	global	0	5	5	5	5	5	5	15
AOW01 Data Ecosystem	High Level Output	HLO 1.8 Governance, incentives for data & digital management culture change	Number of strategies or frameworks adopted to govern and incentivize data and digital management in scientific domains.	Numbers	Report	Report review	Annually	global	0	2	2	2	2	2	2	6
AOW01 Data Ecosystem	Intermediate Outcome	I-OC 1.1 CGIAR and Partners generate FAIR and AI ready data	Number of data-related innovations (e.g., standards, tools, pipelines, repositories) actively adopted by internal or external next users.	Number	Report	Report review	Annually	global	0	2	2	2	2	2	2	6
AOW01 Data Ecosystem	Intermediate Outcome	I-OC 1.2 CGIAR and other organizations mobilize an open, FAIR, secure data and advanced analytics ecosystem	Number of data-related innovations (e.g., standards, tools, pipelines, repositories) actively adopted by internal or external next users.	Number	Report	Report review	Annually	global	0	2	2	2	2	2	2	5
AOW01 Data Ecosystem	Intermediate Outcome	I-OC 1.3 CGIAR and partners capacitated on responsible use of FAIR, AI-ready data	Number of CGIAR centers and partners completing capacity-strengthening programs on responsible use of FAIR and AI-ready data.	Number	Report	Report review	Annually	global	0	5	10	20	20	20	20	95
AOW01 Data Ecosystem	Intermediate Outcome	I-OC 1.4 CGIAR & partners have a culture that drives open, FAIR, AI-ready data & digital resources	Number of institutions that have formally adopted strategies, policies, or incentive mechanisms promoting open, FAIR, and AI-ready data practices.	Number	Report	Report review	Annually	global	0	5	10	10	10	10	10	55
AOW02 Action Lab	High Level Output	HLO 2.1 Research integrates AI analytics and digital technologies to enhance modeling and evidence generation.	Number of research outputs integrated with AI-powered analytics.	Number	Reports	Report review	Annually	global	0	1	1	1	1	1	1	6
AOW02 Action Lab	High Level Output	HLO 2.2 Decision support systems integrate a digital twin approach to enable real-time data use.	Number of decision support systems adopted a digital twin approach for real-time agri-food monitoring.	Numbers	Reports	Report review	Annually	global	0	1	1	1	1	1	1	6
AOW02 Action Lab	High Level Output	HLO 2.3 Breeding programs use AI-powered phenotyping to accelerate trait discovery and breeding cycles.	Number of breeding research outputs integrated with AI-powered analytics.	Numbers	Reports	Report review	Annually	global	0	1	1	1	1	1	1	6
AOW02 Action Lab	High Level Output	HLO 2.4 Genebanks integrate AI-powered tools to improve germplasm management and access.	Number of genebanks deployed AI-powered tools to automate germplasm management and access.	Numbers	Reports	Report review	Annually	global	0	1	1	1	1	1	1	6
AOW02 Action Lab	High Level Output	HLO 2.5 Advisory products use AI language models to improve relevance and user engagement.	Number of advisory services integrate agriculture-focused AI language models.	Number	Report	Report review	Annually	global	0	1	1	1	1	1	1	6
AOW02 Action Lab	High Level Output	HLO 2.6 Capacity sharing programs use AI-powered learning platforms to enhance reach and engagement.	Number of individuals who received capacity-sharing training/courses.	Counting	Training Report	Report review	Annually	global	0	100	200	300	300	200	100	1200
AOW02 Action Lab	High Level Output	HLO 2.7 Digital innovations use human-centered design and citizen science to enhance gender responsiveness and social inclusiveness.	Number of innovations that enhance social inclusiveness.	Number	Report	Report review	Annually	global	0	1	1	1	1	1	1	3
AOW02 Action Lab	Intermediate Outcome	I-OC 2.1 Researchers, practitioners, and policymakers use AI-powered tools and platforms to enhance research productivity, real-time decision-making, and cross-domain collaboration.	Number of CGIAR-supported innovations using AI-powered analytics adopted by researchers, policymakers, and other stakeholders.	Number	Reports	Reports review	Annually	global	0	1	2	2	3	3	3	14
AOW02 Action Lab	Intermediate Outcome	I-OC 2.2 Smallholder farmers, local innovators, women, and youth groups gain equitable access to user-centered, AI-enabled digital services that support learning, innovation, and inclusive participation.	Number of CGIAR-supported innovations (tools and services) innovated by the Accelerator that are being used by the next level of users.	Numbers	Reports and interviews	Report review and interviews	Annually	global	0	1	2	2	1	2	2	10
AOW03 Digital Futures	High Level Output	HLO 3.2 Future case scenarios identified for digital transformation in FVL systems in addressing context-specific challenges and opportunities.	Number of report/other ICT materials produced with Future case scenarios identified for digital transformation in FVL systems in addressing context-specific challenges and opportunities.	Number	Reports	Report review	Annually	global	0	1	1	1	1	1	1	6
AOW03 Digital Futures	Intermediate Outcome	I-OC 3.1 Scientists, CGIAR and partners use a broader range of novel digital technologies to make decisions and investments.	Number of CGIAR centers and/or partners supported in assessing digital solutions for decision-making and investment.	Number	Report	Report review	Annually	global	0	3	7	7	7	7	7	10
AOW03 Digital Futures	Intermediate Outcome	I-OC 3.2 Enhanced capacity of CGIAR and partners to design, develop, and implement inclusive digital solutions consider	Number of digital solutions researched by the AOW3 for next-level users.	Number	Report	Report review	Annually	global	0	3	4	6	6	6	3	30
AOW03 Digital Futures	Intermediate Outcome	I-OC 3.3 Increased awareness and understanding of the potential and risks of frontier digital technologies among key stakeholders.	Number of recommendations adopted to build awareness of frontier digital technology opportunities and risks.	Number	Report	Report review	Annually	global	0	3	5	7	8	8	10	41
AOW04 Enabling Environment	High Level Output	HLO 4.1 Frameworks, methods, and guidelines to catalyze adoption of responsible and inclusive digital and data innovations across FLV systems	Number of Communities of Practices (COPs) on key areas launched and operationalized and maintained	Number	Report	Report review	Annually	global	0	4	6	6	6	6	6	34
AOW04 Enabling Environment	High Level Output	HLO 4.1 Frameworks, methods, and guidelines to catalyze adoption of responsible and inclusive digital and data innovations across FLV systems	The number of framework, methods, guidelines developed by the accelerator to catalyze adoption of responsible and inclusive digital and data innovations across FLV systems	Numbers	Reports	Report review	Annually	global	0	1	2	2	2	2	2	11
AOW04 Enabling Environment	High Level Output	HLO 4.2 Robust, secure and flexible foundational digital infrastructure & sustainable operating model across centers and hubs	Number of new digital tool/enhancements or SOPs developed & deployed or policy/governance changes to enable digital core infrastructure for research and support	Numbers	Report	Report review	Annually	global	0	2	2	2	2	2	2	12
AOW04 Enabling Environment	High Level Output	HLO 4.3 Networks of digital innovation nodes & hubs creating an ecosystem for inclusive and FAIR compliant digital and data innovations	Number of SOPs, knowledge products, reports, scoping studies, new initiatives, and events executed to operationalize the digital innovation nodes.	Number	Report	Report review	Annually	global	0	2	2	2	2	2	2	12
AOW04 Enabling Environment	High Level Output	HLO 4.3 Networks of digital innovation nodes & hubs creating an ecosystem for inclusive and FAIR compliant digital and data innovations	The number of new digital innovation nodes (hubs) created and launched	Numbers	Report	Report review	Annually	global	0	2	3	2	1	1	1	10
AOW04 Enabling Environment	High Level Output	HLO 4.4 Partnership and sustainable business models for digital transformation in FLV systems.	Number of knowledge products on digital innovation, partnerships, and business models for FLV system transformation.	Numbers	Reports	Report review	Annually	global	0	1	1	1	1	1	1	6
AOW04 Enabling Environment	High Level Output	HLO 4.5 Capacity building & training packages on human-centric digital product design, development, sustainability, and business models.	Number of individuals who received training on topics.	Number of individuals	Training report	Training report summary	Annually	global	0	20	50	50	50	50	50	270
AOW04 Enabling Environment	High Level Output	HLO 4.6 AI-based inclusive digital products/innovations co-created in partnership with the digital innovation nodes.	The number of new digital products/innovations co-created with digital innovation nodes.	Numbers	Reports	Report review	Annually	global	0	1	1	2	2	2	2	10
AOW04 Enabling Environment	Intermediate Outcome	I-OC 4.1 Scientists and innovators access methods, tools and policies to generate data that is FAIR and AI	Number of CGIAR-supported tools as well as policies internalized and adopted by partners & organizations.	Number	Report	Report review	Annually	global	0	2	2	2	2	2	2	12
AOW04 Enabling Environment	Intermediate Outcome	I-OC 4.2 Improved capacity across scientists and innovators to develop and deploy responsible digital innovations and sustainable business models.	Number of innovations, including tools and business models, developed through trainings or frameworks.	Numbers	Reports	Report review	Annually	global	0	2	2	2	2	2	2	12
AOW04 Enabling Environment	Intermediate Outcome	I-OC 4.3 Governments adopt policies that promote use of digital innovations in agri-food systems.	Number of policies adopted by decision-makers, including governments and partners.	Number	Reports	Report review	Annually	global	0	1	1	1	1	1	1	6

2. Monitoring plan for HLOs and outcomes

The Monitoring, Evaluation, Learning, and Impact Assessment (MELIA) plan for the Digital Transformation Accelerator (DTA) is anchored in CGIAR's Performance and Results Management Framework (PRMF) and follows guidance from the Portfolio Performance Unit (PPU) and the Portfolio Coordination Unit (PCU). It consists of two integrated components: Monitoring, Evaluation, and Learning (MEL), and Impact Assessment (IA). Together, these components are designed to track progress, enable adaptive management, and evaluate DTA's contributions to CGIAR's broader system-level outcomes.

The DTA's Theory of Change (ToC) includes **23 High-Level Outputs (HLOs)**, which feed into **12 Intermediate Outcomes (I-OCs)**, grouped across four Areas of Work (AoWs). These, in turn, contribute to four defined **2030 Outcomes**. The relationships between HLOs, I-OCs, and the 2030 Outcomes are systematically mapped in the ToC Board, which outlines key result statements, performance indicators, and delivery targets.

As an accelerator, DTA contributes to CGIAR's impact areas indirectly—by enhancing the speed and effectiveness of research, promoting inclusive digital innovation, and supporting the development of a responsible data ecosystem. Therefore, the DTA does not propose standalone impact indicators or independent impact studies. Instead, it contributes to system-level change through collaboration with CGIAR's science programs and other accelerators.

Targets for the HLOs and I-OCs were developed through a consultative process involving discussions between AoW teams and the DTA MELIA focal point. This process ensured that targets are both realistic and aligned with indicator definitions and available evidence on deliverable quality.

Measurement of HLOs is primarily based on **quantitative output indicators**, such as the number of digital tools, platforms, or training events delivered. I-OCs, by contrast, focus on **adoption, integration, and enabling conditions**—including policy uptake, system-level changes, and stakeholder use of innovations. These are also tracked using quantitative indicators but with a stronger emphasis on **influence and uptake**.

The **DTA MELIA focal point** will collaborate closely with **AoW leads** to monitor progress, verify evidence, and report on results, ensuring a coordinated and coherent approach to learning and accountability.

Data Collection Methodology

Given the implementation nature of the DTA, this accelerator will primarily focus on foresight to identify, develop, deploy, and enable an environment around the digital ecosystem. As such, all targets are quantitative. The High-Level Outputs (HLOs) and Intermediate Outcomes (I-OCs) can be directly measured based on the completion and deployment of innovations, as evidenced in completion or annual reports. Furthermore, Key Informant Interviews (KIIs) will be conducted as required to cross-check progress against each HLO and I-OC. These data will be collected on an annual basis for incorporation into the reports. The 2030 Outcomes, are more at adoption-level quantitative indicators and would be measured through reports and /or KIIs as required.

This approach relies on self-reporting from each Area of Work (AoW), where the AoW is responsible for generating reports that document their progress against the defined HLOs, I-OCs, and 2030 Outcomes. Supporting documentation and evidence will be preserved to substantiate the reported progress.

Data Quality Control

As noted in the data collection methodology, due to the delivery nature of the accelerator, DTA will only collaborate with other Science projects, Accelerators, CGIAR Centers, and partners. Therefore, surveys are not required. Since these data are counted on a census basis, no estimation is needed.

Consequently, the data quality strategy will involve:

- **Evidence Preservation:** Ensuring that relevant documentation (e.g., meeting minutes, published guidelines, tool development reports/logs, training attendance sheets, adoption reports, policy documents, etc.) is systematically collected and stored at the MELIA unit. This evidence will serve as primary verification for the reported progress.
- **Reporting Accuracy:** The reports from the AoWs will need to accurately reflect the preserved evidence. This includes an internal review process within each AoW before reports are submitted. KIIs will also be conducted as needed, and their summaries will be recorded.
- **Verification:** This process will involve checking the reported figures against the preserved evidence to ensure accuracy, completeness, and adherence to the defined indicators and targets.
- **Consistency Checks:** Ensuring that the reported numbers align with the definitions of the indicators as provided in the ToC board.

Data Collection Frequency

All data will be collected annually. By the end of every November, all AoW leads will be informed and provided with a template outlining their targets, with a submission deadline at the end of December. The collected data will then be compiled and cross-checked against the provided information. Meetings or Key Informant Interviews (KIIs) will be conducted as required with AoW representatives in early January as part of the quality control process. Following this, the MELIA unit will send the compiled data, including progress and justifications, to the Accelerator Lead for internal approval before incorporation into the PRMS and Type 1 reporting.

3. Evaluation plan

The DTA plans to evaluate its ongoing and deployed interventions through seven distinct studies, as detailed in Annex 2. One foresight study is specifically designed to understand and suggest the future direction of digital technologies within the food, water, and land (FWL) system. Several other studies aim to evaluate the developed tools, helping users identify the most effective options for accelerating their research. Specifically:

- AoW 1 focuses on evaluating the uptake and institutionalization of FAIR and AI-Ready Data Practices within CGIAR.
- AoW 2 investigates the usability and impact of AI language models in agriculture advisory and capacity-sharing services, providing an in-depth understanding of innovation efficiency.
- AoW 3 focuses on foresight and evaluation of innovation.
- AoW 4 will aims on a gender-responsive and inclusive digital transformation strategy.

Except for the foresight study, all other evaluations can be categorized as adoption and diffusion

studies, directly addressing the learning questions outlined in the Theory of Change (ToC). The studies are aligned with the DTA research question as described below:

Area of Work	Research Questions
AoW 1 Data Ecosystems	1.1 How can CGIAR and its partners effectively leverage data and analytics to accelerate research and digital services? 1.2 In what ways can knowledge, attitudes, skills, and habits (KASH) be optimized to deliver AI-ready data and solutions?
AoW 2 Action Lab	2.1. How can AI-powered analytics embedded in agrifood research domains improve the accuracy, timeliness, and usability of decision-support tools for stakeholders? 2.2. Which digital platforms, data systems, and governance models most effectively support the integration of AI into agrifood systems risk management? 2.3. How do inclusive design practices influence equitable access to, adoption of, and impact AI-powered digital tools among small-scale producers, women, youth, and local innovators?
AoW 3 Digital Future	3.1 How will digital transformation impact FLW systems in developing countries over the next decade and what are the potential socioeconomic and environmental implications? 3.2 How can policy frameworks & innovation strategies harness emerging digital frontier technologies in agri-food systems while ensuring equitable access and environmental sustainability in developing regions?
AoW 4 Enabling environment	4.1 Can organization innovations including a network of innovations hubs and cross-center COEs lead to an increase in the development and use of inclusive and ethical data and digital innovations in FLW systems 4.2 What policy innovations, governance, and business models institutionalize FAIR & AI-ready ecosystems, inclusive and ethical digital innovation processes?

Evaluation studies

A list of studies with relevant information is provided in Annex 2.

Evaluation collaboration plans

The Digital Transformation Accelerator (DTA) will collaborate with fourteen bilateral initiatives and projects, extending the reach and impact of its innovations and services through these partnerships. This collaboration enables the DTA to scale its contributions across a wider set of stakeholders and contexts by leveraging the work and platforms of other projects.

However, as of 2025—during the inception phase—detailed information about the collaboration strategies and mechanisms is still emerging. Therefore, it is premature to conduct a formal evaluation of these collaborations at this stage. As partnerships evolve and more information becomes available on how joint implementation is progressing, a collaborative evaluation plan will be developed in a future iteration of the MELIA framework.

To illustrate the current scope of collaboration, a table is provided below, showing the alignment between DTA's High-Level Outputs (HLOs) and the specific bilateral projects. This mapping

demonstrates how DTA's innovations are expected to complement and contribute to the outcomes of these bilateral initiatives.

Sl#	W3/Bilateral project name	Center	Area of Work Title and Number	DTA HLO
1.	T-PJ-003866-Liberia Land and Soil Resources Knowledge project (Soils4Liberia)	IITA	AoW2 Action Lab	HLO 2.1.1
2.	N-600317-STATS FR SPACE-MOZAMBIQUE	IFPRI	AoW2 Action Lab	HLO 2.1.1
3.	N-343004-AgPile	IFPRI	AoW1 Data Ecosystem	HLO 1.2.4
4.	R-A-2024-19-ISARC Phase III	IRRI	AoW2 Action Lab	HLO 2.1.2
5.	T-PJ-003763-IITA Accelerated Innovation Delivery Initiative	IITA	AoW4 Enabling Environment	HLO 4.1.1
6.	F-AG10607-ASIA-AFRICA BLUETECH SUPERHIGHWAY PROJECT (COAST)	WorldFish	AoW4 Enabling Environment	HLO 4.1.1
7.	T-PJ-003942-IITA Accelerated Innovation Delivery Initiative (AID-I) for Mozambique	IITA	AoW4 Enabling Environment	HLO 4.1.1
8.	N-343005-GAIA Phase II	IFPRI	AoW2 Action Lab	HLO 2.1.5
9.	PEPSI-Climate-Resilient Agriculture/Thailand	ABC	AoW2 Action Lab	HLO 2.1.1
10	W-D-0523-DIWASA - Phase II	IWMI	AoW2 Action Lab	HLO 2.1.2
11	Bill & Melinda Gates Foundation(BMGF)-1000 Farms Research Platform	ABC	AoW2 Action Lab	HLO 1.1.2
12	W-D-0392-NET: WaPOR Phase 2	IWMI	AoW2 Action Lab	HLO 2.1.2
13	Climate-smart initiatives for climate change adaptation and sustainability in prioritized agricultural production systems in Colombia (CSICAP)	ABC	AoW3 Digital Futures	HLO 3.1.1
14	USA-BMGF-Artemis II - Field Phenotyping Prototype	ABC	AoW2 Action Lab	HLO 2.1.3

4. Impact assessment

Program/Accelerator Impact Overview

At this **inception phase**, the Digital Transformation Accelerator (DTA) is **not planning to conduct internal impact assessments**. This decision reflects the early stage of implementation, where systems, partnerships, and innovations are still being established, and sufficient data may not yet be available to meaningfully assess long-term outcomes or impacts.

However, as implementation progresses and more evidence becomes available, **opportunities to explore the impact of DTA's investments** may emerge. These could include examining how DTA's digital innovations, tools, and approaches are contributing to CGIAR's system-wide outcomes—such as improved decision-making, enhanced productivity, and greater inclusivity in digital access and use.

In the meantime, the DTA remains fully open and committed to collaborating on impact assessments initiated or recommended by CGIAR's overarching frameworks. Specifically, DTA will coordinate with assessments aligned to the Performance and Results Management Framework (PRMF) and follow guidance provided by the Portfolio Performance Unit (PPU) and the Portfolio Coordination Unit (PCU). This ensures that any impact assessment activities are strategically aligned with CGIAR's system-wide priorities and contribute to broader learning and accountability.

In future iterations of the MELIA plan, as implementation deepens and use cases mature, DTA will explore the feasibility of targeted studies that assess its contribution to the uptake, effectiveness, and scaling of digital innovations across CGIAR and its partners. These studies will be designed in coordination with MELIA stakeholders to ensure relevance, quality, and strategic value.

IA activities and collaboration plans

To be provided in the next iteration.

IA studies

To be provided in the next iteration.

5. Learning and adaptive management

The Accelerator team will document, share, and make use of lessons learned for continuous improvement. Adaptive management is based on both operational process-related learning, making use of lessons as they arise, and research-based learning, making use of lessons in a periodic reflective process that focuses on revisiting the Accelerator ToC.

Learning

In 2025, the Digital Transformation Accelerator (DTA) envisions conducting **four internal MELIA learning studies**. These studies are designed to examine the **adoption and diffusion of digital innovations** and to **anticipate the need for novel solutions** across food, land, and water (FLW) systems.

The focus will be on generating actionable insights into how innovations are being integrated, the factors that enable or hinder their uptake, and where emerging demands signal the need for new tools or approaches. These findings will support **evidence-based adaptation and prioritization**

of future innovations within the DTA.

The learnings from these studies will not only inform internal planning but will also be shared with other CGIAR **science programs and accelerators**. By offering practical insights into the conditions and strategies that support successful innovation uptake, the studies will help other programs make informed **decisions on digital investments** that can accelerate their research and impact.

In addition, these learning studies will feed into DTA's broader contribution to **cross-program knowledge sharing** within CGIAR, positioning the Accelerator as both a driver and a facilitator of digital transformation. The studies will be documented and disseminated through briefs, presentations, and learning sessions to ensure accessibility and uptake across CGIAR and partner organizations.

Annual review and adaptive planning

Will incorporate it in the upcoming iteration.

6. Reporting

Reporting is a critical function within the Digital Transformation Accelerator (DTA), serving both internal and external communication needs. It ensures accountability, transparency, and effective progress tracking, while also enabling data-driven decision-making, supporting adaptive management, and showcasing DTA's achievements to stakeholders.

DTA will deliver two key types of reports:

Narrative Reporting (Type 1)

This report will capture qualitative progress, lessons learned, implementation challenges, and contextual insights. It provides a comprehensive view of activities, performance, and learning across the Accelerator.

PRMS Reporting

DTA will also submit structured reports through the Performance and Results Management System (PRMS), following the instructions and reporting timelines provided by the Portfolio Performance Unit (PPU) and the Portfolio Coordination Unit (PCU).

To support these processes, relevant information from the DTA Results Framework, including High-Level Outputs, Intermediate Outcomes, and 2030 Outcomes, will be integrated into the PRMS reporting tool via the PRMS reporting options. This will enable accurate reporting on both output and outcome indicators, ensuring alignment with CGIAR system-level results.

Moreover, reporting will be supported by links to MELIA studies and evidence sources, ensuring that reported achievements are substantiated with credible data and insights.

Together, these reporting mechanisms will contribute to a coherent and robust system of performance communication and accountability, strengthening DTA's value to both CGIAR and its broader community of stakeholders.

7. MELIA resources

MELIA budget and staff

The Monitoring, Evaluation, Learning, and Impact Assessment (MELIA) function within the Digital Transformation Accelerator (DTA) will be coordinated by a dedicated **MELIA Focal Point**. This role is central to ensuring alignment with CGIAR's performance frameworks and facilitating results-based learning and adaptive management across the Accelerator. A MELIA budget is provided in Annex 3.

Key Responsibilities of the MELIA Focal Point:

- **Coordination with System-Level Units:** Serve as the primary liaison with CGIAR's Performance and Results Management Framework (PRMF) and follow the guidance provided by the Portfolio Performance Unit (PPU) and the Portfolio Coordination Unit (PCU) to ensure DTA's MELIA efforts are aligned with broader CGIAR standards and processes.
- **Support to AoW Leads:** Collaborate closely with Area of Work (AoW) leads to clarify and track their MELIA-related commitments. This includes regular communication of committed targets and timelines to ensure smooth and consistent implementation across all components of the Accelerator.
- **Data Collection and Reporting:** Lead the timely collection, consolidation, and submission of MELIA data. This ensures progress against indicators (HLOs and I-OCs) is accurately captured and reported in line with CGIAR requirements.
- **Adaptive Management:** Analyze findings and share insights with the Accelerator Core Team to inform strategic adjustments, improve delivery, and respond to emerging challenges and opportunities during implementation.
- **Learning and Knowledge Sharing:** Identify and document key successes, challenges, and lessons learned throughout the MELIA cycle engaging all the Area of Works. These insights will be used to inform future iterations of the DTA and contribute to organizational learning across CGIAR.

8. Data management

General Approach

Data management plan

The Digital Transformation Accelerator (DTA) is committed to responsible, open, and secure data management in alignment with the CGIAR Open and FAIR Data Assets Policy. This plan outlines the approach to storing, managing, and sharing data generated through DTA activities, ensuring data is accessible, useful, and protected throughout its lifecycle.

- **Data Classification:** DTA will categorize data into three types: Open, Restricted, and Internal, based on accessibility and sensitivity. Each dataset will be tagged accordingly to guide storage and sharing protocols.
- **Data Storage:** If required, data will be stored in secure, backed-up environments using trusted CGIAR or partner infrastructures. File formats will be chosen for long-term preservation, and storage will support version control and redundancy.
- **Metadata and Documentation:** All datasets will be accompanied by standardized metadata to ensure they are findable and understandable, especially data curated under AoW1. Metadata will follow standards and include key details like origin, purpose, and licensing.
- **Data Sharing:** DTA will publish open data through CGIAR-approved repositories or global platforms as a mandate of AoW1 with compliance and ethical obligations.
- **Legal and Ethical Compliance:** All data activities will comply with data protection laws (like PII), CGIAR policies, and ethical research practices. Sensitive data will be anonymized, and necessary permissions will be secured before sharing.

- **Data Standard:** Throughout the data collection, curation, analysis, and reporting process the best standard process will be ensured to ensure the quality of the submitted report.
- **Monitoring and Improvement:** DTA will regularly assess and update its data management practices. Training will be provided to ensure all team members understand and apply CGIAR data principles effectively.

Annex 2. MELIA studies template

MELIA Study title	MELIA type	Lead Center	Contact	HLO, I-OC, 2030 OC	Indicator(s) evidenced by this study	Location	Methods and design approaches	Start date for the study	End date for the study	How related to past evaluations (SG or others)
AI Co-Scientist to accelerate agriculture research and development	Adoption or diffusion studies addressing learning questions on the TOC	International Water Management Institute	• Garcia Andarcia Mariangel (m.garciaandarcia@cgiar.org)	HLO 3.1: I-OC 3.1.I-OC 3.2. I-OC 3.3.	• Number of probes of a wide range of AI and/or other novel digital solutions to support system-level scenario analysis of digital frontier technology impacts for decision-makers	Global	Consultation workshops, Literature review, experts' interview, development of probes.	2025	2025	Water Copilot
Evaluating the Uptake and Institutionalization of FAIR, AI-Ready Data Practices in CGIAR	Adoption or diffusion studies addressing learning questions on the TOC	International Maize and Wheat Improvement Center / Centro Internacional de Mejoramiento de Maíz y Trigo	• Andrea Gardeazabal (a.gardeazabal@cgiar.org)	I-OC 1.1 - I-OC1.2 - I-OC1.3 I-OC1.4		Global	Mixed methods: document review, platform analytics, pre/post training surveys, key informant interviews, and institutional policy audits.	2028	2028	Big data platform report
Evaluating the Usability and Impact of AI Language Models in Agricultural Advisory and Capacity Sharing Services	Qualitative outcome study	International Food Policy Research Institute	• Jawoo Koo (j.koo@cgiar.org)	I-OC 2.2. HLO 2.5.	• Number of CGIAR-supported innovations (tools and services) innovated by the Accelerator that are being used by the next level of users.	Global	User surveys, A/B testing, focus group interviews	2026	2027	N/A
Exploratory Probes of Frontier Digital Technologies for Strategic Decision Support in CGIAR	Adoption or diffusion studies addressing learning questions on the TOC	International Water Management Institute	• Garcia Andarcia Mariangel (m.garciaandarcia@cgiar.org)	I-OC 3.2. I-OC 3.3.- I-OC 3.1.HLO 3.1s	• This indicator will count the number of CGIAR centers and/or partners supported it by assessing their existing/newly deployed digital solutions to make decisions and investments.	Global		2025	2025	Virtual Reality, Meta Humans, and digital twins.
Gender responsive and inclusive digital transformation strategy	Adoption or diffusion studies addressing learning questions on the TOC	International Rice Research Institute	• Niyati Singaraju (n.singaraju@cgiar.org)	I-OC 4.1, 2030 OC 2		Global		2025	2025	Builds on prior consultation workshops conducted as part of the work package on digital inclusion under the CGIAR initiative on Digital Innovation/ and other evidence summaries and reports consolidated by the CGIAR GENDER Impact Platform
Strategic Foresight for the Digital Futures of FLW Systems	Ex-ante, baseline and/or foresight study	Alliance of Bioversity and CIAT - Headquarter (Bioversity International)	• Jimenez Daniel (d.jimenez@cgiar.org)	I-OC 3.3.		Global	Scenario planning workshops, strategic foresight methods, futures thinking, Opinion pieces, Design thinking,	2025	2025	N/A

MELIA study type is given for reference in **Annex 4.**

Annex 3. MELIA budget templates

MELIA Budget: Staff Cost Breakdown by Year

Instructions: Replicate this reporting table for each year from 2025 to 2030 to plan staff and input costs accordingly. Additionally, you may add rows as necessary to cover each activity and staff type per activity.

Activity title	Activity description	Staff (# per type)	Full-time equivalent (# days)	Staff cost (USD)	Inputs i.e. travel costs, special equipment costs	Inputs cost (USD)	Total cost (USD)
Monitoring activities: Cross-cutting							
<i>Activities related to the routine monitoring of HLOs and outcomes</i>							
		#, type					N/A
		#, type					
MELIA studies: Cross-cutting							
<i>MELIA studies to evidence the achievement of program-level impacts and outcomes</i>							
		#, type					N/A
		#, type					
MELIA studies: AoW outcomes							
<i>MELIA studies to evidence the achievement of AoW-level intermediate outcomes not covered by any cross-cutting MELIA studies</i>							
AoW 1- Data Ecosystems	MELIA Study	01					250,000
AoW 2- Action Lab	MELIA Study	01					150,000
AoW 3- Digital Futures	MELIA Studies	03					TBD
AoW 4- Action Lab	MELIA Study	01					40,000
Total MELIA activity costs							

Next steps: Use the total cost column for each year to fill in the **MELIA Budget: Total Cost** table below.

MELIA Budget: Total Cost

Activity title	Activity description	2025–2030 Total cost (USD)					
		2025 (Budget)	2026 (Forecast)	2027 (Forecast)	2028 (Forecast)	2029 (Forecast)	2030 (Forecast)
1. Monitoring activities: Cross-cutting							
Activities related to the routine monitoring of HLOs and outcomes							
N/A							
2. MELIA studies: Cross-cutting							
MELIA studies to evidence the achievement of program-level impacts and outcomes							
N/A							
Sub-total: Cross-cutting MELIA costs							
3. MELIA studies: AoW outcomes							
MELIA studies to evidence the achievement of AoW-level intermediate outcomes not covered by any cross-cutting MELIA studies							
MELIA studies		40,000	TBD	250,000	150,000	-	-
Sub-total: Other MELIA costs				250,000	150,000		
TOTAL: All MELIA costs				250,000	150,000		

Next steps: Activities and costs entered under sections 1 and 2 (captured by the cross-cutting sub-total) should be aggregated under the “cross-cutting” section of the **Plan of Results and Budget (PORB) template**, shared separately. Activities and costs entered under section 3 should be entered under the relevant AoW outcome in the PORB template.

Annex 4. MELIA study types

For reference only, the current MELIA study types defined in the [2024 Technical Reporting Guidance](#) are listed below. An updated MELIA studies typology will be shared in 2025.

Type	Definition
Causal Impact Assessment learning studies	These are research studies designed to test key assumptions underlying different steps of the causal chain (theory of change) that links CGIAR research/innovations to high-level impacts. As learning studies, they can be implemented in a 3-year cycle and use credible counterfactuals to provide early evidence of causal impacts of intermediary outcomes in the theory of change, and effectiveness of different approaches/interventions to improve such outcomes. This additional credible evidence becomes a relevant feedback to the research process that aims to maximize the impact of CGIAR related innovations in the long-term.
Adoption or diffusion studies addressing learning questions on the TOC	A study that identifies the spread, acquisition and use of social, institutional or technological innovations. Adoption and diffusion studies can address learning questions within an Initiative's theory/ies of change by documenting whether and how innovation/s have reached intended beneficiaries. Assessments of adoption/acquisition/use seek to identify whether innovation/s have been taken up or rejected by intended beneficiaries, in order to make a case for CGIAR contribution to outcome/s, relative to other potential influencing factors. Together, studies that encompass diffusion and adoption assessments show how innovations have spread to and been adopted or rejected by end users. Note: Adoption and diffusion studies do not necessarily assess impact. However, impact assessments may, and often do, include assessments of diffusion and adoption.
Tracing of scaling activities and policy advice, as a base for long-term, large scale impact assessments	LTLS impact studies usually require a longtime frame to observe high-level impacts associated to the use of CGIAR related innovations or policy recommendations. The design of these studies should be done from the start when CGIAR related innovations are ready to initiate the scaling process. The tracing activities should be designed in order to document how, when, where and why CGIAR related innovations and policy advice are disseminated or made to next users.
Qualitative outcome study	A study that assesses changes in behaviors, practices, perceptions or attitudes among beneficiaries of an innovation, or those who can facilitate the diffusion and adoption of an innovation (e.g., policymakers). Qualitative outcome studies are often used to substantiate contributions to policies, investments, budgets, curricula or similar.
Program/project evaluation or review	A program/project evaluation refers to a systematic and objective assessment of an on-going or completed program or project. Program evaluations focus on the evaluation of a set of time-bound interventions involving multiple activities that may cut across sectors, themes and/or geographic areas. These would also include evaluations of Initiatives. Project evaluations focus on the evaluation of an individual intervention designed to achieve specific objectives within specified resources and implementation schedules, often within the framework of a broader program. Evaluations are external, completely or largely independent, and systematic studies of an in-depth nature using clear evaluation criteria, whereas reviews may be more flexible and narrower in focus.
Ex-ante, baseline and/or foresight study	An ex-ante study, also known as a baseline study, is conducted before an intervention to determine the baseline conditions against which future change, outcomes and impact can be assessed. A foresight study involves the structured and explicit exploration of multiple futures in order to inform decision-making. Foresight studies are usually conducted at the beginning of an intervention, but may be used throughout the duration of a project or program to refine decision- and priority-making.
Scaling readiness assessment	An assessment of how "ready" innovations are for scaling and the appropriate actions for the acceleration and/or enhancement of scaling. Scaling readiness assessments can also identify which scaling bottlenecks need to be addressed, the most cost-effective scaling strategies, and which partners to engage.
Other MELIA activity (please specify)	A study or analysis related to monitoring, evaluation, learning or impact assessment that does not belong to the abovementioned types. The study or analysis will in some way test assumptions, inform learning and adaptive management, meet accountability requirements and/or inform the design of new Initiatives, programs and projects. When choosing "other MELIA activity", you must justify how the study or analysis you are reporting constitutes a MELIA activity. Examples that could be added here include monitoring and synthesis studies.